Double Descent in Linear Models

Assignment 4

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1 Introduction

Classical statistical learning theory predicts a U-shaped test error curve, adding model complexity first reduces bias, but eventually increases variance, so there is a sweet spot. This bias-variance trade-off has guided model selection and regularization for decades.

Recent empirical findings reveal an important gap. In high-dimensional, overparameterized settings, the test error can exhibit a *second descent* after the interpolation threshold, the point at which the model perfectly fits training data. This **double descent** phenomenon challenges classical intuition and helps explain why large, overparameterized neural networks can still generalize well. Belkin et al. [1] provide a systematic study in linear regression, which is reproduced here.

The goal of this assignment is to empirically demonstrate double descent, understand its mechanism via weight-norm analysis, and study the effect of regularization (Ridge), noise level, and optimization method (gradient descent vs. closed form).

2 Dataset

We use a synthetic regression dataset with noisy linear labels:

$$w^* = \mathbf{1}_d/\sqrt{d}, \qquad X \sim \mathcal{N}(0, I_{n\times d}), \qquad y = Xw^* + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n). \tag{1}$$

The core experiment fixes $n = 100$ training samples and sweeps the feature dimension $d \in \{10, 15, \ldots, 300\}$, so the ratio $\gamma = d/n$ ranges from 0.1 to 3.0, covering the under-parameterized part, the interpolation threshold $d = n$, and well into the over-parameterized part. A test set of $n_{\text{test}} = 2000$ samples is drawn independently for evaluation.

Design choices. Setting $w^* = \mathbf{1}/\sqrt{d}$ keeps $\|w^*\|_2 = 1$ for all d, so the Bayes-optimal risk is constant and comparisons across dimensions are fair. Each (d, trial) pair uses a deterministic seed for full reproducibility, and 30 independent trials per point provide mean ± std bands in all plots. Three noise levels $\sigma \in \{0.1, 0.5, 1.0\}$ are used in the noise-sweep experiments.

3 Implementation

All algorithms are implemented from scratch in Python using only NumPy for array operations and Matplotlib for visualization. The code is organized into the following modules:

Table 1: Code modules and their responsibilities.

Module	**Purpose**
`data.py`	Synthetic dataset generation
`ols.py`	Minimum-norm OLS solver (normal equations & dual form)
`ridge.py`	Ridge regression solver
`solvers.py`	Gaussian elimination, Householder SVD, Moore–Penrose pseudo-inverse
`metrics.py`	MSE and weight-norm computation
`experiments.py`	OLS/Ridge/GD sweep orchestration
`plotting.py`	All figure-generation functions

3.1 OLS

The objective $\min_w \|Xw-y\|_2^2$ is solved in two branches. For $d < n$ (over-determined), the normal equations $\hat{w} = (X^\top X)^{-1}X^\top y$ give a unique solution. For $d \geq n$ (under-determined), infinitely many solutions exist; we select the **minimum-norm** one via the dual form $\hat{w} = X^\top(XX^\top)^{-1}y$, This is equivalent to the Moore–Penrose pseudo-inverse $\hat{w} = X^+y$. The key insight is that solving the $n \times n$ system $K\alpha = y$ (where $K = XX^\top$) is cheaper than working with the full $d \times d$ parameter space, and it automatically selects the solution of minimum ℓ_2 norm.

3.2 Ridge

Adding a penalty $\lambda\|w\|_2^2$ yields closed-form solutions in the same two branches:

$$d \leq n: \quad \hat{w}_\lambda = (X^\top X + \lambda I_d)^{-1}X^\top y, \qquad d > n: \quad \hat{w}_\lambda = X^\top(XX^\top + \lambda I_n)^{-1}y.$$

The λI term makes the system strictly positive definite for any $\lambda > 0$, eliminating the singularity at $d = n$.

3.3 Custom linear algebra (`solvers.py`)

To demonstrate the mechanics from scratch, the following primitives are implemented without calling any high-level solver:

`solve(A, b)` Gaussian elimination with back-substitution (no partial pivoting). Solves $Aw = b$ for square A.

`_bidiagonalise(A)` Reduces $A \in \mathbb{R}^{m\times n}$ $(m \geq n)$ to upper bidiagonal form $B = U_b A V_b$ using successive Householder reflections applied from the left (to zero out sub-diagonal entries) and from the right (to zero out super-diagonal entries beyond the first superdiagonal).

`_householder_svd(A)` Computes the full SVD $A = U\Sigma V^\top$ by bidiagonalizing A and then applying `numpy`'s SVD on the small bidiagonal matrix.

`pinv(A)` Computes the Moore–Penrose pseudo-inverse $A^+ = V\Sigma^+U^\top$, where singular values below a threshold $\tau = \max(m, n) \cdot \epsilon_{\text{mach}} \cdot \sigma_1$ are replaced by zero.

`fit_weights(X, y)` Combines the above: normal equations for $d < n$, pseudo-inverse otherwise. Numerical tests confirm errors of order 10^{-16} relative to `numpy.linalg.solve` and `numpy.linalg.pinv`.

3.4 Gradient descent

Gradient descent for the MSE loss with step size η follows:

$$w_{t+1} = w_t - \frac{\eta}{n}X^\top(Xw_t - y), \quad w_0 = \mathbf{0}.$$

Starting from $w_0 = \mathbf{0}$, GD is implicitly biased toward the component of the solution in the row space of X, which is exactly the minimum-norm solution. Consequently, GD with sufficient iterations should converge to the same solution as the closed-form pseudo-inverse.

4 Results

All experiments use $n = 100$ training samples and 30 independent trials per (d, λ) pair unless stated otherwise., and MSE clipped at 20 for visual clarity (the spike near $d = n$ can be arbitrarily large in individual trials).

4.1 OLS train and test error: double descent

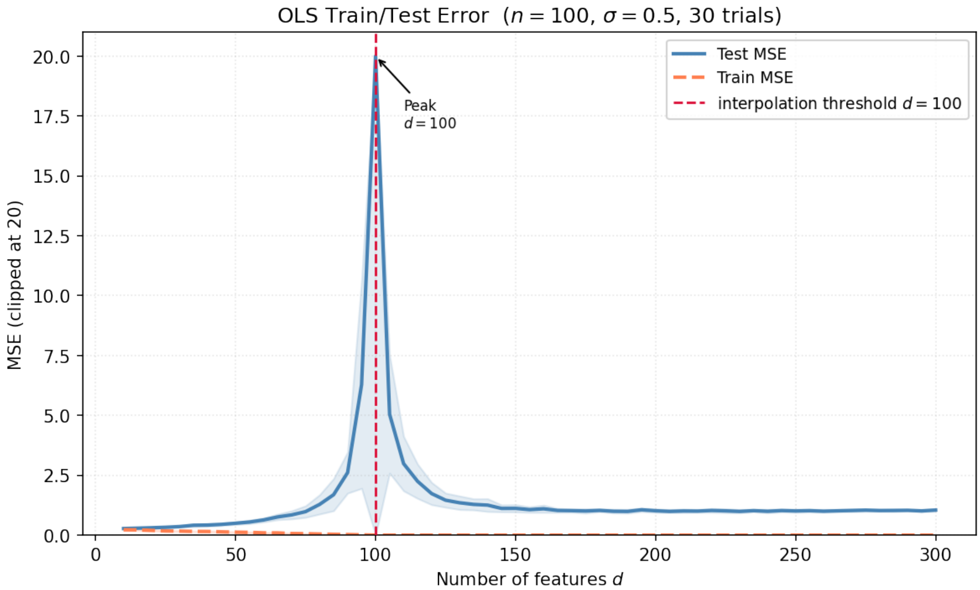

Figure 1: OLS train and test MSE vs. feature dimension d ($n = 100$, $\sigma = 0.5$, 30 trials).

Figure 1 clearly demonstrates double descent. The test error traces the double descent shape:

1. **First descent** ($d < n$): as more features are added, the model can explain more of the signal, reducing underfitting. Test MSE falls.
2. **Peak** ($d \approx n$): the system is barely determined; $XX^\top$ has its smallest eigenvalue near zero for Gaussian random matrices, and inverting this near-singular matrix catastrophically amplifies noise. The test error spikes.
3. **Second descent** ($d > n$): the minimum-norm pseudo-inverse solution distributes the fitted weights across all d features. Each weight coefficient shrinks as d grows, so the model generalizes progressively better.

4.2 OLS vs. Ridge regression

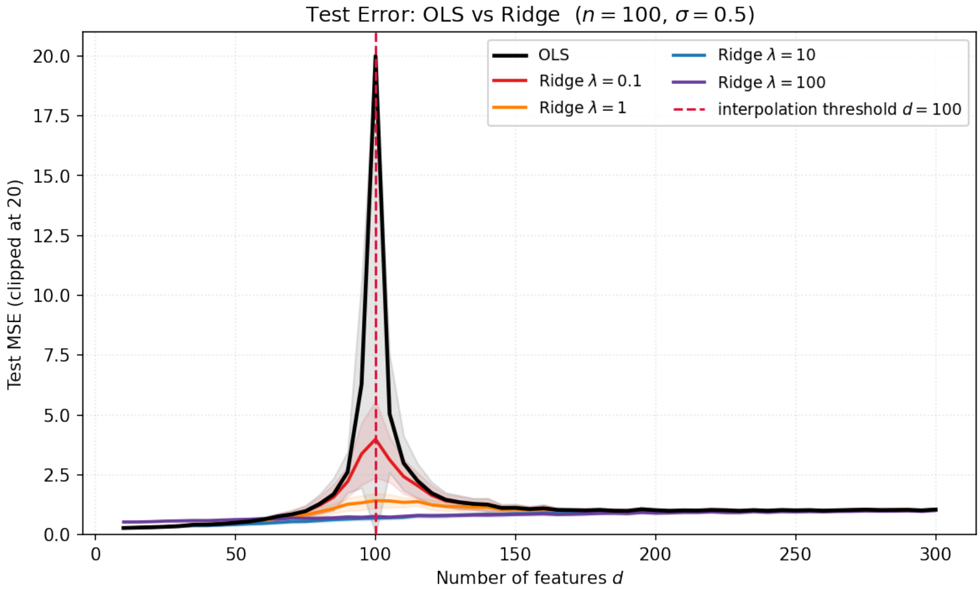

Figure 2: Test MSE for OLS and Ridge with $\lambda \in \{0.1, 1, 10, 100\}$ ($n = 100$, $\sigma = 0.5$).

Figure 2 shows that Ridge regression eliminates the interpolation-threshold peak entirely. Since $(XX^\top + \lambda I)$ is strictly positive definite for any $\lambda > 0$, the near-singularity that causes the spike is removed. Smaller λ (red, orange) closely approximates OLS away from the threshold while providing a modest smoothing near it; larger λ (blue, purple) is stable everywhere but introduces substantial bias, raising the error in both regimes. An optimally tuned λ matching the noise level outperforms OLS everywhere, though selecting it requires cross-validation or prior knowledge of σ.

4.3 Effect of noise

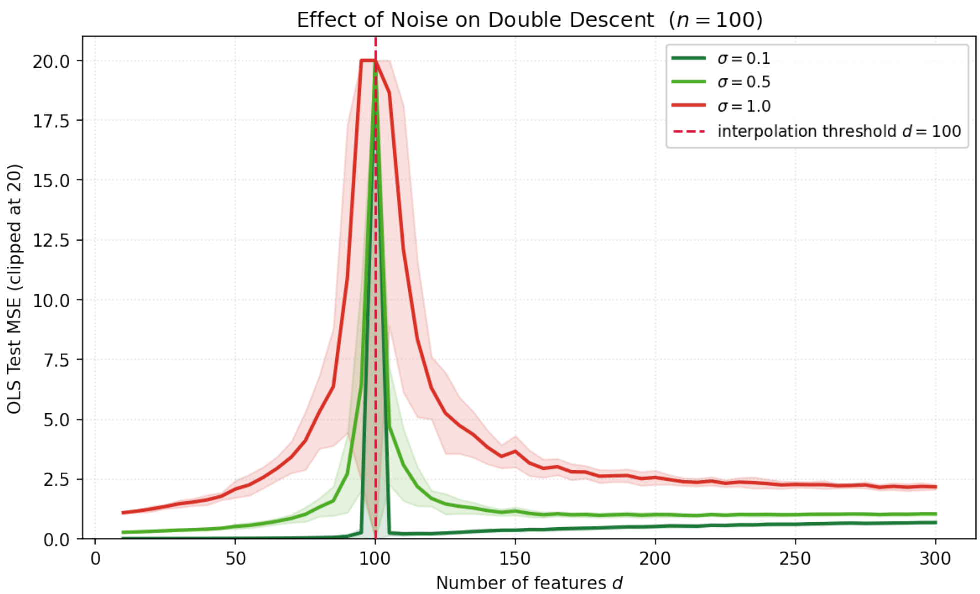

Figure 3: OLS test MSE for $\sigma \in \{0.1, 0.5, 1.0\}$ ($n = 100$, 30 trials).

Figure 3 shows how noise affects the double descent curve. The overall shape is similar for all noise levels. However, higher noise (σ) makes the peak wider. With high noise, the test error starts to increase much earlier, even before reaching $d = n$. Also, after the peak ($d > n$), higher noise makes the recovery slower. This means the model needs many more features to handle the noise and become stable again.

4.4 Weight norm analysis

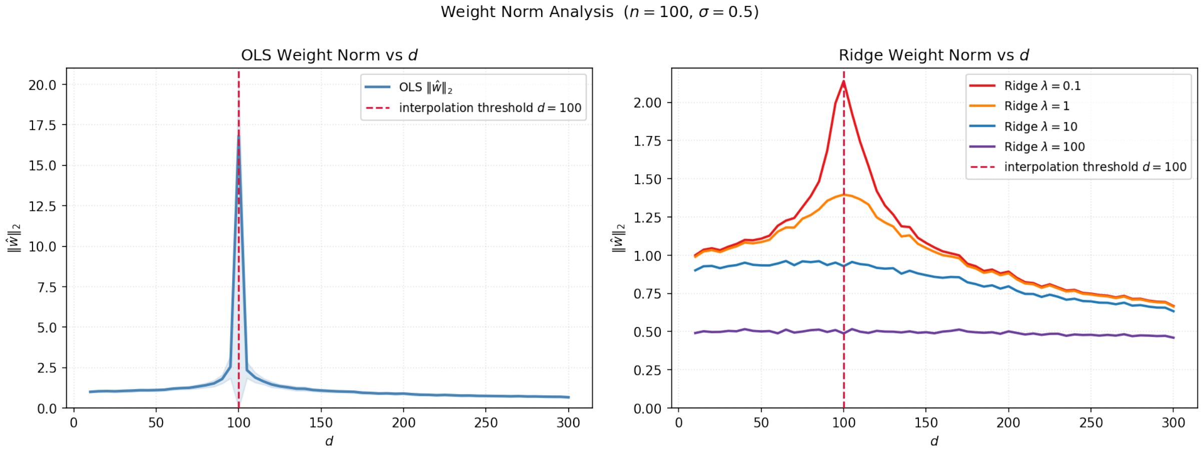

Figure 4: ℓ_2 weight norm for OLS (left) and Ridge (right) vs. d ($n = 100$, $\sigma = 0.5$).

Figure 4 provides the mechanistic explanation for the double descent peak.

OLS norm: The OLS weight norm (left panel) spikes sharply at $d \approx n$, exactly where the test MSE also spikes. For $d < n$, the normal-equations solution has moderate norm. At $d = n$, the smallest eigenvalue of $X^\top X$ is close to zero, so inverting it produces a weight vector with enormous components that absorb noise. For $d > n$, the minimum-norm pseudo-inverse spreads the weight across all d features; each component shrinks as $d/n \to \infty$, which explains why the test error recovers: small-norm weights generalize better.

Ridge norm control: The Ridge norms (right panel) are monotonically controlled: larger λ yields smaller norms for all d. Crucially, the norms are bounded everywhere and do not spike at $d = n$, because the λI term prevents the matrix inversion from amplifying noise. This directly links Ridge regularization to the suppression of the double descent peak.

The correspondence between the norm blow-up and the test-error spike is one of the most illuminating results of this study: the double descent peak is caused by the minimum-norm OLS solution needing very large coefficients to interpolate noisy data when the system is barely determined.

4.5 Gradient descent vs. closed-form OLS

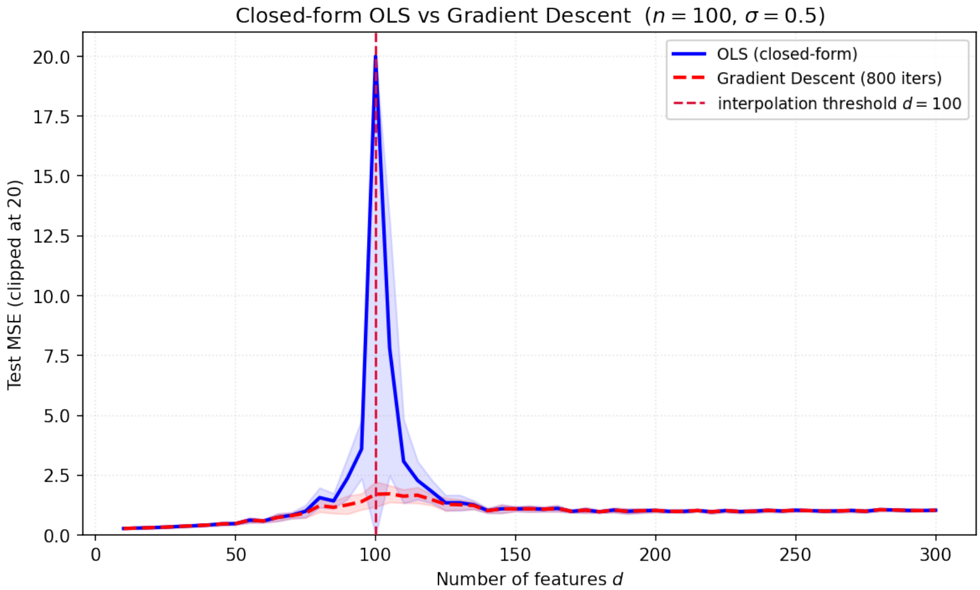

Figure 5: Closed-form OLS vs. gradient descent (800 epochs, $\eta = 0.1$) ($n = 100$, $\sigma = 0.5$, 10 trials).

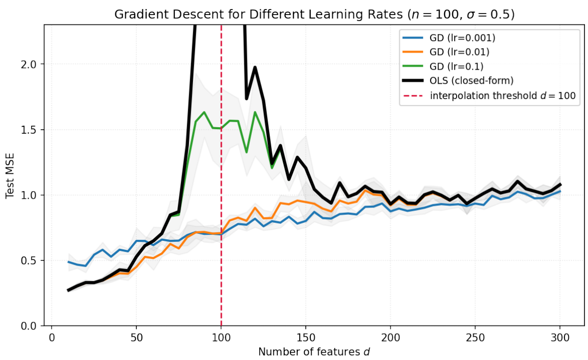

Figure 6: Effect of learning rate (η) on GD test MSE. Fixed 800 epochs.

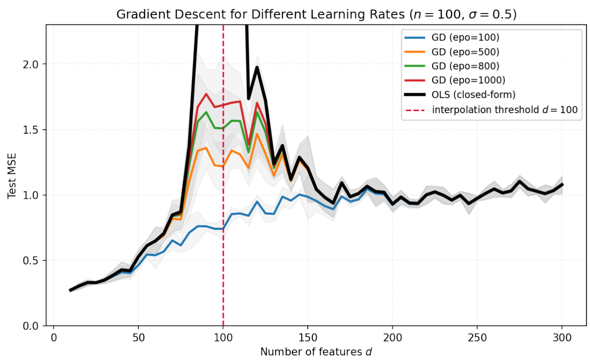

Figure 7: Effect of number of epochs on GD test MSE. Fixed $\eta = 0.1$.

Figure 5 shows that GD with 800 epochs and $\eta = 0.1$ closely reproduces the closed-form OLS curve, including the double descent shape, confirming the implicit minimum-norm bias of zero-initialized gradient descent.

Figure 6 shows the effect of learning rate: $\eta = 0.001$ is too slow to converge in the overparameterized regime, while $\eta = 0.1$ matches closed-form OLS well. Near the threshold, even moderate rates can be unstable as the condition number of $X^\top X/n$ diverges.

Figure 7 reveals the most practically important result: with only 100 epochs, GD has not yet reached the interpolating solution, its weight norm stays small, and the test-error peak is substantially suppressed. As training continues the peak emerges, and at 1000 epochs the curve nearly matches the closed-form solution. **Early stopping thus acts as implicit Ridge regularization**, limiting weight-norm growth in the same way that $\lambda > 0$ does. This provides a principled explanation for why early stopping helps in overparameterized settings.

5 Discussion and Conclusion

The experiments cleanly reproduce all the qualitative features described by Belkin et al. in a controlled linear setting. The peak at the interpolation threshold is clearly visible; Ridge regression systematically smooths it; and the weight-norm analysis provides a mechanistic explanation. The gradient-descent experiments add the practically relevant insight that early stopping acts as implicit Ridge regularization.

The use of multiple trials (30) and large test sets ($n_{\text{test}} = 2000$) ensures that the results are statistically stable. The error bands in the figures give a clear picture of both the mean behavior and the variability, especially important near the threshold where single-trial fluctuations can be enormous.

This project empirically reproduced the double descent phenomenon in linear regression. The key findings are summarized below.

Table 2: Summary of experimental findings.

Experiment	**Main finding**
OLS train/test sweep	Clear double descent: test MSE peaks sharply at $d = n$ then recovers monotonically in the over-parameterized regime.
OLS vs. Ridge	Ridge eliminates the peak by preventing the weight norm from diverging; optimal λ outperforms OLS everywhere.
Noise sweep	The double descent shape is universal; peak height and the second-descent floor both scale with σ^2.
Weight norm	The test-error peak is mechanistically caused by the norm blow-up of the minimum-norm OLS solution at $d \approx n$.
Gradient descent	GD from zero converges to the minimum-norm solution; early stopping acts as implicit Ridge regularization and smooths the curve.

References

[1] M. Belkin, D. Hsu, S. Ma, and S. Mandal. Reconciling modern machine learning practice and the classical bias-variance trade-off. *PNAS*, 116(32):15849–15854, 2019.